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Display apparatus and compensating method of image data

Abstract

A compensating method of image data is provided. The compensating method includes: calculating a polarity accumulation time of a positive polarity and a negative polarity of the image data; and determining whether to compensate a voltage polarity of the image data according to the polarity accumulation time.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6232944	12/2000	Kumagawa	345/94	G09G 3/3696
8907936	12/2013	Mizusako	N/A	N/A
9343017	12/2015	Yamada	N/A	N/A
9697783	12/2016	Yeo	N/A	G09G 3/3614
10276109	12/2018	Crounse et al.	N/A	N/A
10732444	12/2019	Chida	N/A	N/A
2008/0309656	12/2007	Van Woudenberg	345/87	G09G 3/3614
2010/0042866	12/2009	Shih	713/600	G06F 1/12
2010/0073341	12/2009	Toyooka	345/55	G09G 3/3614
2010/0309183	12/2009	Mizusako	N/A	N/A
2011/0216100	12/2010	Yamada	N/A	N/A
2014/0085345	12/2013	Ma	345/89	G09G 3/3655
2017/0263175	12/2016	Crounse et al.	N/A	N/A
2017/0358255	12/2016	Wang	N/A	G09G 3/2055
2019/0228717	12/2018	Crounse et al.	N/A	N/A
2019/0310502	12/2018	Chida	N/A	N/A
2020/0175930	12/2019	Koo	N/A	G09G 3/3614
2021/0327327	12/2020	Fujikawa	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101908302	12/2009	CN	N/A
102194415	12/2010	CN	N/A
109074781	12/2017	CN	N/A
110121743	12/2018	CN	N/A
113763901	12/2020	CN	N/A

OTHER PUBLICATIONS

“Office Action of Taiwan Counterpart Application”, issued on Apr. 16, 2025, p. 1-p. 5. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the priority benefit of U.S. Provisional application Ser. No. 63/557,580, filed on Feb. 25, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(1) The invention relates to an electronic device and a compensating method, more specifically, to a display apparatus and a compensating method of an image data.

Description of Related Art

(2) In a normal display, each pixel of a liquid crystal display (LCD) panel should switch between positive and negative frames. However, if the LCD panels stay in a single polarity for a long period, it may result in impurity ions in the LCD panels interfering with the electric field under the influence of direct current (DC) bias, thereby causing flicker.

(3) The timing of the LCD apparatus is defined by the front-end system. The timing controller is part of the receiving end. In the normal display, the timing controller receives timing information from the front-end system. In a scenario where even frames are longer than odd frames, the LCD apparatus will flicker more, and vice versa.

SUMMARY

(4) The invention is directed to a display apparatus and a compensating method of an image data, capable of improving display quality.

(5) The invention provides a compensating method of image data including: calculating a polarity accumulation time of a positive polarity and a negative polarity of the image data; and determining whether to compensate a voltage polarity of the image data according to the polarity accumulation time.

(6) The invention provides a display apparatus including a display panel and a driver circuit. The display panel is configured to display images according to image data. The driver circuit is coupled to the display panel. The driver circuit is configured to output the image data to drive the display panel to display the images. The driver circuit is further configured to calculate a polarity accumulation time of a positive polarity and a negative polarity of the image data, and determine whether to compensate a voltage polarity of the image data according to the polarity accumulation time.

(7) To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

(2) FIG. 1 is a schematic block diagram illustrating a display apparatus according to an embodiment of the invention.

(3) FIG. 2 is a waveform diagram of image data according to an embodiment of the invention.

(4) FIG. 3 is a flowchart illustrating steps in the compensating method of the image data according to an embodiment of the invention.

(5) FIG. 4 is a schematic diagram illustrating a polarity accumulation time changing along with image frames according to an embodiment of the invention.

DESCRIPTION OF THE EMBODIMENTS

(6) Embodiments are provided below to describe the disclosure in detail, though the disclosure is not limited to the provided embodiments, and the provided embodiments can be suitably combined. The term “coupling/coupled” or “connecting/connected” used in this specification (including claims) of the application may refer to any direct or indirect connection means. For example, “a

first device is coupled to a second device” should be interpreted as “the first device is directly connected to the second device” or “the first device is indirectly connected to the second device through other devices or connection means.” In addition, the term “signal” can refer to a current, a voltage, a charge, a temperature, data, electromagnetic wave or any one or multiple signals.

(7) FIG. 1 is a schematic block diagram illustrating a display apparatus according to an embodiment of the invention. FIG. 2 is a waveform diagram of image data according to an embodiment of the invention. Referring to FIG. 1 and FIG. 2, the display apparatus **100** includes a driver circuit **110** and a display panel **120**. The display panel **120** is configured to display images according to the image data VD. The driver circuit **110** is coupled to the display panel **120**. The driver circuit **110** is configured to output the image data VD to drive the display panel **120** to display the images in a plurality of image frames F1 to F6. The waveform diagram of the image data VD is illustrated in FIG. 2. The image data VD is an image data for driving a specified pixel of display panel **120**.

(8) Each of the image frames F1 to F6 includes an active period VA and a blanking period VB. The image data VD has a voltage of a positive polarity in the image frames F1, F3 and F5. The image data VD has a voltage of a negative polarity in the image frames F2, F4 and F6. The polarities of the voltage alternatively distribute in the image frames F1 to F6. In the present embodiment, the image frames F1, F3 and F5 with the positive polarity is short than the image frames F2, F4 and F6 with the negative polarity, but the invention is not limited thereto. In other embodiments, the image frames F1, F3 and F5 with the positive polarity may be longer than or equal to the image frames F2, F4 and F6 with the negative polarity.

(9) In the present embodiment, the driver circuit **110** may be a display driver integrated circuit (DDIC), and include a timing controller and a source driver integrated in a single circuit chip, but the invention is not limited thereto. In other embodiments, the timing controller and the source driver may be disposed in different driver circuits. In addition, enough teaching, suggestion, and implementation illustration for the hardware structures of the driver circuit **110** and the display panel **120** can be obtained with reference to common knowledge in the related art, which is not repeated hereinafter.

(10) In the disclosure, the driver circuit **110** may perform a compensating method on the image data VD to avoid flicker of the display panel **120**. To be specific, FIG. 3 is a flowchart illustrating steps in the compensating method of the image data according to an embodiment of the invention. FIG. 4 is a schematic diagram illustrating a polarity accumulation time changing along with image frames according to an embodiment of the invention.

(11) Referring to FIG. 3 and FIG. 4, the image frames F1, F3, F5, F7, F10, F13 and F15 marked as plus sign “+” indicate that these image frames have the positive polarity (hereinafter “the positive frames”), and the image frames F2, F4, F6, F8, F9, F11, F12, F14 and F16 marked as minus sign “-” indicate that these image frames have the negative polarity (hereinafter “the negative frames”). The time length of the positive frames F1, F3, F5 and F7 before compensation is 16.6 milliseconds, and the time length of the negative frames F2, F4, F6 and F8 before compensation is 8.3 ms. The time length of the frames is taken for example, and does not intend to limit the invention.

(12) In step S100, the driver circuit **110** calculates a polarity accumulation time $t_{_ACC}$ of the positive polarity and the negative polarity of the image data VD every even frames before compensation. In FIG. 4, the driver circuit **110**, for example, calculates the polarity accumulation time $t_{_ACC}$ every four frames before compensation, but the invention is not limited thereto. For example, the positive frames F1 and F3 each have a time length of 16.6 ms, and the negative frames F2 and F4 each have a time length of 8.3 ms. For accumulation calculation, the time length of the negative frames F2 and F4 is considered as a negative value, and thus at point **401**, the polarity accumulation time $t_{_ACC}$ is calculated as 16.6 ms. Therefore, the polarity accumulation time $t_{_ACC}$ indicates a time difference of the positive polarity and the negative polarity of the image data VD.

(13) In step **S110**, the driver circuit **110** determines whether to compensate a voltage polarity of the image data VD according to the polarity accumulation time t_{ACC} . When the polarity accumulation time t_{ACC} is smaller than or equal to the first threshold value t_{TH1} , the driver circuit **110** does not compensate the voltage polarity of the image data VD. Alternatively, when the polarity accumulation time t_{ACC} is larger than or equal to the second threshold value t_{TH2} , the driver circuit **110** also does not compensate the voltage polarity of the image data VD, wherein the second threshold value t_{TH2} is smaller than the first threshold value t_{TH1} . In this case, the first threshold value t_{TH1} is set as 20 ms, and the second threshold value t_{TH1} is set as -20 ms, but the invention is not limited thereto.

(14) At point **401**, since the polarity accumulation time t_{ACC} is calculated as 16.6 ms and smaller than the first threshold value t_{TH1} , i.e. $16.6\text{ ms} < 20\text{ ms}$, the driver circuit **110** determines not to compensate the voltage polarity of the image data VD. The driver circuit **110** returns to step **S100** and continues to calculate the polarity accumulation time t_{ACC} every four frames. The polarity accumulation time t_{ACC} is not reset.

(15) Next, at point **402**, the polarity accumulation time t_{ACC} is calculated as 33.2 ms, which is larger than the first threshold value t_{TH1} . When the polarity accumulation time t_{ACC} is larger than the first threshold value t_{TH1} , the driver circuit **110** compensates the voltage polarity of the image data VD in a first polarity pattern. Alternatively, when the polarity accumulation time t_{ACC} is smaller than a second threshold value t_{TH2} , the driver circuit **110** compensates the voltage polarity of the image data VD in a second polarity pattern.

(16) Table 1 shows the polarity distribution of the first polarity pattern and the second polarity pattern.

(17) TABLE-US-00001 TABLE 1 polarity pattern polarity distribution $t_{ACC} > t_{TH1}$ first polarity pattern $-+-$ $+-$ $---$ $---$ $t_{ACC} < t_{TH2}$ second polarity pattern $+-$ $+-$ $+-$ $+-$

(18) The expression $t_{ACC} > t_{TH1}$ indicates that the polarity accumulation time t_{ACC} of the positive frames is too long, and thus, the negative polarity compensation is required. The first polarity pattern is adopted for the negative polarity compensation. The first polarity pattern includes at least two frames of the negative polarity.

(19) The expression $t_{ACC} < t_{TH2}$ indicates that the polarity accumulation time t_{ACC} of the negative frames is too long, and thus, the positive polarity compensation is required. The second polarity pattern is adopted for the positive polarity compensation. The second polarity pattern includes at least two frames of the positive polarity.

(20) In the disclosure, each of the first polarity pattern and the second polarity pattern has time length of odd frames. Table 1 shows each of the first polarity pattern and the second polarity pattern has time length of three frames as an example, and the invention is not limited thereto.

(21) At point **402**, since the polarity accumulation time t_{ACC} is calculated as 33.2 ms and larger than the first threshold value t_{TH1} , i.e. $33.2\text{ ms} > 20\text{ ms}$, the driver circuit **110** determines to compensate the voltage polarity of the image data VD. The driver circuit **110** performs step **S120** and compensates the voltage polarity of the image data VD in the first polarity pattern.

(22) In step **S120**, the driver circuit **110** selects one polarity distribution “ $-+-$ ” of the first polarity pattern **410** to compensate the voltage polarity of the image data VD. As illustrated in FIG. 4, starting from point **402**, the voltage polarity of the image data VD is compensated using the first polarity pattern **410**. The frames F9, F10 and F11 are respectively the negative frame, the positive frame and the negative frame, and have the same time length 8.3 ms, but the invention is not limited thereto.

(23) After compensation, the driver circuit **110** goes to step **S130**, and calculates the polarity accumulation time t_{ACC} every odd frames, e.g. every three frames in this case. At point **403**, since the polarity accumulation time t_{ACC} is calculated as 24.9 ms, and still larger than the first threshold value t_{TH1} , i.e. $24.9\text{ ms} > 20\text{ ms}$, the driver circuit **110** performs step **S120** again, and compensates the voltage polarity of the image data VD in the first polarity pattern **420**.

(24) Next, at point **404**, the polarity accumulation time t_{ACC} is calculated as 16.6 ms, and smaller than the first threshold value t_{TH1} , i.e. $16.6\text{ ms} < 20\text{ ms}$. The driver circuit **110** stops compensating the voltage polarity of the image data VD, and returns to step **S100** from step **S110**. That is to say, when the polarity accumulation time t_{ACC} is between the first threshold value t_{TH1} and the second threshold value t_{TH2} , the driver circuit **110** stops compensating the voltage polarity of the image data VD. In addition, the polarity accumulation time t_{ACC} is calculated every odd frames after compensation until the compensation is ended.

(25) In summary, in the embodiments of the invention, the driver circuit calculates the polarity accumulation time every even frames, and sets threshold values for compensation. When the polarity compensation is required, the driver circuit compensates the voltage polarity of the image data in the specified polarity pattern, and calculates the polarity accumulation time every odd frames until the compensation is ended. Consequently, the voltage polarity of the image data is compensated in order to prevent the occurrence of flicker on the display panel. This results in the display apparatus being able to provide a high-quality display.

(26) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. A compensating method of image data, comprising: calculating a polarity accumulation time of a positive polarity and a negative polarity of the image data; and determining whether to compensate a voltage polarity of the image data according to the polarity accumulation time, wherein the step of determining whether to compensate the voltage polarity of the image data according to the polarity accumulation time comprising: when the polarity accumulation time is larger than a first threshold value, compensating the voltage polarity of the image data in a first polarity pattern; and when the polarity accumulation time is smaller than a second threshold value, compensating the voltage polarity of the image data in a second polarity pattern, wherein the second threshold value is smaller than the first threshold value.
2. The compensating method of the image data according to claim 1, wherein each of the first polarity pattern and the second polarity pattern has time length of odd frames.
3. The compensating method of the image data according to claim 2, wherein the first polarity pattern comprises at least two frames of the negative polarity.
4. The compensating method of the image data according to claim 2, wherein the second polarity pattern comprises at least two frames of the positive polarity.
5. The compensating method of the image data according to claim 1, further comprising: when the polarity accumulation time is between the first threshold value and the second threshold value, stop compensating the voltage polarity of the image data.
6. The compensating method of the image data according to claim 1, wherein the polarity accumulation time of the positive polarity and the negative polarity of the image data is calculated every even frames before compensation.
7. The compensating method of the image data according to claim 1, wherein the polarity accumulation time of the positive polarity and the negative polarity of the image data is calculated every odd frames after compensation until the compensation is ended.
8. The compensating method of the image data according to claim 1, wherein the step of determining whether to compensate the voltage polarity of the image data according to the polarity accumulation time comprising: when the polarity accumulation time is smaller than or equal to the first threshold value, not compensating the voltage polarity of the image data; and when the polarity accumulation time is larger than or equal to the second threshold value, not compensating

the voltage polarity of the image data.

9. The compensating method of the image data according to claim 1, wherein the polarity accumulation time indicates a time difference of the positive polarity and the negative polarity of the image data.

10. A display apparatus, comprising: a display panel, configured to display images according to image data; and a driver circuit, coupled to the display panel, and configured to output the image data to drive the display panel to display the images, wherein the driver circuit is further configured to calculate a polarity accumulation time of a positive polarity and a negative polarity of the image data, and determine whether to compensate a voltage polarity of the image data according to the polarity accumulation time, wherein when the polarity accumulation time is larger than a first threshold value, the driver circuit compensates the voltage polarity of the image data in a first polarity pattern; and when the polarity accumulation time is smaller than a second threshold value, the driver circuit compensates the voltage polarity of the image data in a second polarity pattern, wherein the second threshold value is smaller than the first threshold value.

11. The display apparatus according to claim 10, wherein each of the first polarity pattern and the second polarity pattern has time length of odd frames.

12. The display apparatus according to claim 11, wherein the first polarity pattern comprises at least two frames of the negative polarity.

13. The display apparatus according to claim 11, wherein the second polarity pattern comprises at least two frames of the positive polarity.

14. The display apparatus according to claim 10, wherein when the polarity accumulation time is between the first threshold value and the second threshold value, the driver circuit stops compensating the voltage polarity of the image data.

15. The display apparatus according to claim 10, wherein the driver circuit calculates the polarity accumulation time of the positive polarity and the negative polarity of the image data every even frames before compensation.

16. The display apparatus according to claim 10, wherein the driver circuit calculates the polarity accumulation time of the positive polarity and the negative polarity of the image data every odd frames after compensation until the compensation is ended.

17. The display apparatus according to claim 10, wherein when the polarity accumulation time is smaller than or equal to the first threshold value, the driver circuit does not compensate the voltage polarity of the image data; and when the polarity accumulation time is larger than or equal to the second threshold value, the driver circuit does not compensate the voltage polarity of the image data.

18. The display apparatus according to claim 10, wherein the polarity accumulation time indicates a time difference of the positive polarity and the negative polarity of the image data.
